Complete HTTP Reference

Message Format, Methods, Status Codes, and Headers

1 HTTP Message Format

Each HTTP message is text-based and consists of four parts:

- **Start Line** – request line (client) or status line (server).
- **Header Lines** – metadata about the request/response.
- **Blank Line** – separates headers from body.
- **Message Body** – optional data such as HTML, JSON, files.

1.1 Request Fields

Common Request Headers

Header	Meaning
Host	Domain name of the server (required in HTTP/1.1).
User-Agent	Identifies client software (e.g., browser, app).
Accept	Media types client can handle (e.g., text/html, application/json).
Accept-Language	Preferred languages (e.g., en-US).
Accept-Encoding	Compression formats client supports (gzip, deflate).
Authorization	Credentials for authentication (e.g., Basic, Bearer token).
If-Modified-Since	Requests resource only if it has changed since given date.
Cookie	Sends stored cookies to server.
Referer	Page that linked to the resource.

1.2 Response Fields

Common Response Headers

Header	Meaning
Server	Identifies the server software (e.g., Apache, Nginx).
Date	Timestamp when response was sent.
Content-Type	Type of returned data (e.g., text/html, application/json).
Content-Length	Size of response body in bytes.
Content-Encoding	Compression method used (gzip, br).
Set-Cookie	Sends cookies from server to client.
Location	Redirect URL (used with 3xx codes).
Cache-Control	Caching policies (e.g., no-cache, max-age=3600).
ETag	Unique identifier for version of resource.

2 HTTP Methods

HTTP Methods Overview

Method	Description
GET	Retrieve information (safe, cacheable).
POST	Submit data (form submissions, uploads). Not idempotent.
PUT	Upload or replace a resource. Idempotent.
DELETE	Remove a resource. Idempotent.
HEAD	Same as GET but only headers returned.
OPTIONS	Describes supported methods and features.
TRACE	Diagnostic loopback (returns request as received).
CONNECT	Establish tunnel (commonly for HTTPS).
PATCH	Apply partial modifications to a resource.

Properties:

- **Safe:** methods that do not modify resources (GET, HEAD, OPTIONS, TRACE).
- **Idempotent:** multiple identical requests have same effect (GET, PUT, DELETE, HEAD, OPTIONS).
- **Cacheable:** responses can be stored (GET, HEAD, sometimes POST).

3 HTTP Status Codes

Status codes are grouped into five categories:

1. **1xx Informational** – Request received, continuing process.

2. **2xx Success** – Request successfully processed.
3. **3xx Redirection** – Further action needed to complete request.
4. **4xx Client Error** – Error on client side.
5. **5xx Server Error** – Error on server side.

3.1 1xx Informational

100	Continue	Initial request accepted, client should continue.
101	Switching Protocols	Server switching protocols (e.g., HTTP/1.1 to HTTP/2).
103	Early Hints	Allows preloading resources.

3.2 2xx Success

200	OK	Request succeeded.
201	Created	Resource successfully created.
202	Accepted	Request accepted, processing later.
204	No Content	Success but no body returned.
206	Partial Content	Used with range requests.

3.3 3xx Redirection

301	Moved Permanently	Resource moved permanently.
302	Found	Temporary redirect.
303	See Other	Redirect to another URI with GET.
304	Not Modified	Resource unchanged since last request.
307	Temporary Redirect	Temporary redirect (method unchanged).
308	Permanent Redirect	Permanent redirect (method unchanged).

3.4 4xx Client Error

400	Bad Request	Invalid request syntax.
401	Unauthorized	Authentication required.
403	Forbidden	Server refuses request.
404	Not Found	Resource not available.
405	Method Not Allowed	Method not supported for resource.
408	Request Timeout	Client took too long.
429	Too Many Requests	Rate limit exceeded.

3.5 5xx Server Error

500	Internal Server Error	Generic server error.
501	Not Implemented	Server does not support method.
502	Bad Gateway	Invalid response from upstream server.
503	Service Unavailable	Server overloaded or down.
504	Gateway Timeout	No response from upstream server.
505	HTTP Version Not Supported	Unsupported HTTP version.